
DIGITAL DINING APPLICATION FOR HOTELS**Atharva Amar Vadam^{*1}**^{*1}Dr D.Y Patil, Polytechnic, Kolhappur, India.

I. INTRODUCTION

This digital dining application aims to transform the traditional dining experience by integrating technology directly at the dinner table. With the convenience of this app, hotel guests no longer have to wait for a server to take their order or ask for assistance. By simply using a tablet at the table or their own mobile device, guests can explore an interactive digital menu with detailed images, descriptions, and prices, making it easy to select and customize their meals.

The app is designed with a user-friendly interface, optimized for intuitive navigation, and equipped with features that ensure orders are delivered accurately and promptly. Additionally, the application provides real-time updates on the status of the order, allowing guests to know exactly when their meal is being prepared or served. This enhances the overall dining experience by combining convenience, customization, and speed. For the hotel, this application helps reduce dependency on staff for repetitive tasks, lowers the chance of human error, and allows employees to focus more on providing exceptional hospitality. It also opens opportunities to gather insights on popular dishes, peak order times, and customer preferences, which can inform menu adjustments and service improvements.

The **Digital Menu Card for Hotel** is an innovative application designed to enhance the guest experience by replacing traditional paper menus with an interactive digital solution. This system will allow customers to view, select, and place orders directly from a digital device (such as tablets or smartphones). The main goal is to improve the efficiency of the ordering process, streamline restaurant operations, and offer an intuitive, eco-friendly, and modern dining experience. As part of this project, we are using **HTML**, **CSS**, and **JavaScript** to build the front-end of the application, ensuring that the interface is user-friendly, responsive, and efficient. This web application is designed to be accessed via a browser, making it compatible with various devices such as smartphones, tablets, and laptops.

Working**1. Login and Access to Digital Menu :**

- Guests can scan a QR code on their table or log in to the app using their room number and a secure pin.
- Once logged in, the app displays a visually engaging digital menu. It includes various categories like starters, main courses, desserts, and beverages, making it easy to navigate.
- The menu also showcases daily specials, chef recommendations, and dietary options (like vegan or gluten-free), all in an attractive, photo-rich format.

2. Customizable Ordering Process :

- Guests can add items to their cart, adjust quantities, or add notes for each dish (e.g., "extra spicy" or "no onions").
- They can also view calorie counts, ingredients, and allergen information, helping them make informed choices
- Once selections are finalized, they review the cart and submit their order. The app may prompt for confirmation to ensure accuracy.

3. Real-Time Order Tracking :

- After placing the order, guests can see a confirmation screen with an estimated wait time.
- The app provides real-time status updates, such as "Order Received," "In Preparation," and "Ready for Delivery."
- Notifications are sent to guests' devices when the order is on its way to the table, allowing them to anticipate and prepare for its arrival.

4. Flexible Payment Options :

- The app keeps a running total of the bill, which guests can view at any time. This transparency helps guests manage their dining expenses.
- Payment options include direct in-app payment via credit/debit card, mobile wallets, or linking to their hotel room account for checkout upon departure.
- Guests can choose to split the bill, add a tip, and request a digital or printed receipt.

5. Additional Features and Custom Requests:

- Guests can use the app to make requests beyond food, such as asking for additional napkins, refills, or other table service needs.
- The app has a feedback section where guests can rate each item and provide comments on their experience.
- Loyalty features and personalized recommendations based on guest preferences can be added, encouraging repeated engagement and offering a tailored experience.

6. Integration with Hotel Systems :

- The app is designed to integrate seamlessly with the hotel's existing kitchen display systems (KDS) and POS (point-of-sale) systems, ensuring efficient order processing and payment handling.
- Data analytics from the app help hotel management track sales trends, item popularity, and customer preferences. This data can drive strategic decisions for menu updates and marketing.

II. LITERATURE SURVEY

Related work

Research on digital ordering systems in hospitality has grown, driven by the demand for contactless and efficient customer service. Similar systems are widely implemented in fast-food chains, where customers use self-service kiosks to place orders, enhancing speed and order accuracy. Applications like UberEATS and Grubhub have pioneered mobile food ordering platforms, which have increased the expectation for digital solutions in dining. Studies show that digital ordering systems can significantly reduce order wait times and improve customer satisfaction by allowing customization options and providing real-time order tracking. Hotel-based solutions have focused primarily on in-room dining apps but are now expanding into restaurant settings to improve the dine-in experience.

In recent years, the hospitality industry has seen significant advancements in the way food services are managed and delivered, especially with the rise of **digital dining applications**. These systems are transforming the dining experience by allowing customers to interact with menus, place orders, and even make payments through their mobile devices, reducing the need for traditional methods of interaction such as physical menus or waitstaff.

Existing work

In traditional hotel dining settings, guests typically order through waitstaff, who manually take orders at each table. Some restaurants use handheld devices or tablets for servers, enabling them to enter orders directly into a POS system, which is then sent to the kitchen. While these systems improve order accuracy and efficiency for the staff, they still require servers as intermediaries. A few hotels have introduced basic digital menus through QR codes, but they often lack interactivity, customization, or real-time tracking, limiting the effectiveness and functionality for guests.

Problems of existing system

Existing systems present several challenges, particularly in providing a seamless and efficient ordering experience:

1. **Dependence on Servers** : Guests must rely on waitstaff to place and manage orders, which can lead to delays, especially during peak hours or staff shortages.
2. **Order Errors** : Manual order-taking introduces a risk of errors in order details, leading to poor customer satisfaction and increased food waste.
3. **Lack of Real-Time Updates** : Traditional ordering doesn't allow guests to track their order's status, which

often results in longer perceived wait times.

4. **Limited Menu Information** : Printed or static digital menus often lack details such as allergen information, calorie counts, and customization options.
5. **Payment Inflexibility** : Most existing systems rely on physical payment methods, which can be inconvenient and limit flexibility.

Problem description

To improve the dining experience in hotel restaurants, a need exists for a digital application that enables guests to order directly from their table, reducing the reliance on servers and minimizing order errors. The goal is to provide a streamlined, interactive, and user-friendly platform that enhances the traditional dining process by making it faster, more convenient, and transparent for guests. By implementing real-time updates, order tracking, and flexible payment options, the application aims to address the inefficiencies of current dining systems in hotel settings.

Goal & Objective

Goal : Develop a comprehensive digital ordering system for hotel restaurants that allows guests to order food directly from their dinner table, enhancing the overall dining experience.

Objectives :

1. **Enable Direct Ordering** : Allow guests to place orders directly from their table using a digital menu on a tablet or smartphone.
2. **Increase Order Accuracy** : Provide detailed customization options for guests, minimizing the potential for errors in the order.
3. **Provide Real-Time Order Updates** : Offer real-time notifications on order status, from preparation to delivery, keeping guests informed.
4. **Offer Flexible Payment Options** : Integrate various payment methods, including in- app payment and room billing, for greater convenience.
5. **Improve Efficiency and Guest Satisfaction** : Streamline the ordering process, reduce wait times, and create a more enjoyable dining experience.

Application

The proposed digital dining application has several key applications:

1. **Hotel Restaurants** : Improve the guest experience by enabling convenient, self- service ordering, particularly during busy times when waitstaff are occupied.
2. **Food Courts and Buffet Restaurants** : Enhance operations in larger, self-service settings by allowing guests to order select dishes directly from the table.
3. **In-Room Dining** : Extend the application to support in-room dining, allowing guests to order from anywhere within the hotel.
4. **Analytics for Hotel Management** : Provide data on ordering trends, peak dining hours, and popular items, helping hotel managers make informed decisions on menu offerings and staffing.
5. **Personalized Dining Experience** : Allow hotels to provide personalized recommendations based on guest order history and preferences, further enhancing guest satisfaction.

III. METHODOLOGY

Proposed methodology

The methodology for developing the Digital Dining Application involves several stages, from understanding user requirements to final testing and deployment. Below is a breakdown of each step:

- **Requirement Gathering and Analysis**

1. Conduct surveys and interviews with stakeholders (hotel managers, restaurant staff, and guests) to gather detailed requirements.
2. Identify user needs, such as the ability to browse menus, customize orders, and make payments seamlessly.
3. Determine non-functional requirements such as system scalability, security, and responsiveness.

• Development of Key Features

1. Menu Display and Interaction: Users will be able to view the digital menu with categories, images, prices, and descriptions.
2. Order Cart Functionality: Users can add, remove, and update items in their cart. The total price is dynamically calculated using JavaScript.
3. Checkout and Payment: Develop a mock payment system that simulates the checkout process.
4. Order History: If implemented, the system can track user orders and show a history of past orders.

• Testing and Quality Assurance

1. Conduct functionality testing, usability testing, and performance testing to ensure the system works smoothly.
2. Gather feedback from users to identify pain points and improve the design.

Problem description

In many hotel and restaurant environments, the dining experience is still largely dependent on traditional paper menus and in-person interactions with waitstaff. This leads to delays in order placement, potential misunderstandings, and less efficient service. In addition, the COVID-19 pandemic accelerated the need for contactless solutions, creating a demand for digital systems that allow customers to order food, view menus, and make payments without physical interactions. This project aims to address these problems by developing a Digital Dining Application that enables hotel guests to view digital menus, place orders, and make payments directly from their mobile devices.

System Design Documentation

The system design documentation outlines the architecture and components of the Digital Dining Application. It covers both the front-end and back-end design (even if the back-end is simulated in this case) and provides a clear roadmap for building and testing the application.

Frontend Design:

1. **Menu Page** : Displays the list of food items with descriptions, images, and prices. This page is styled using CSS and allows users to interact with the menu via JavaScript for adding items to their cart.
2. **Cart Page** : Displays a summary of selected items, quantities, and total price. Users can edit their cart before proceeding to checkout.
3. **Checkout Page** : A mock payment form where users can "pay" for their order (simulated transaction).
4. **Responsive Layout** : Uses CSS media queries to adapt the design to various screen sizes (mobile, tablet, desktop).

Backend (Simulated) :

1. **Order Management System** : Simulates the process of receiving and processing orders. This includes updating the order status, generating an order ID, and calculating totals.
2. **Payment Processing** : Simulated transaction using JavaScript to show a mock payment confirmation.

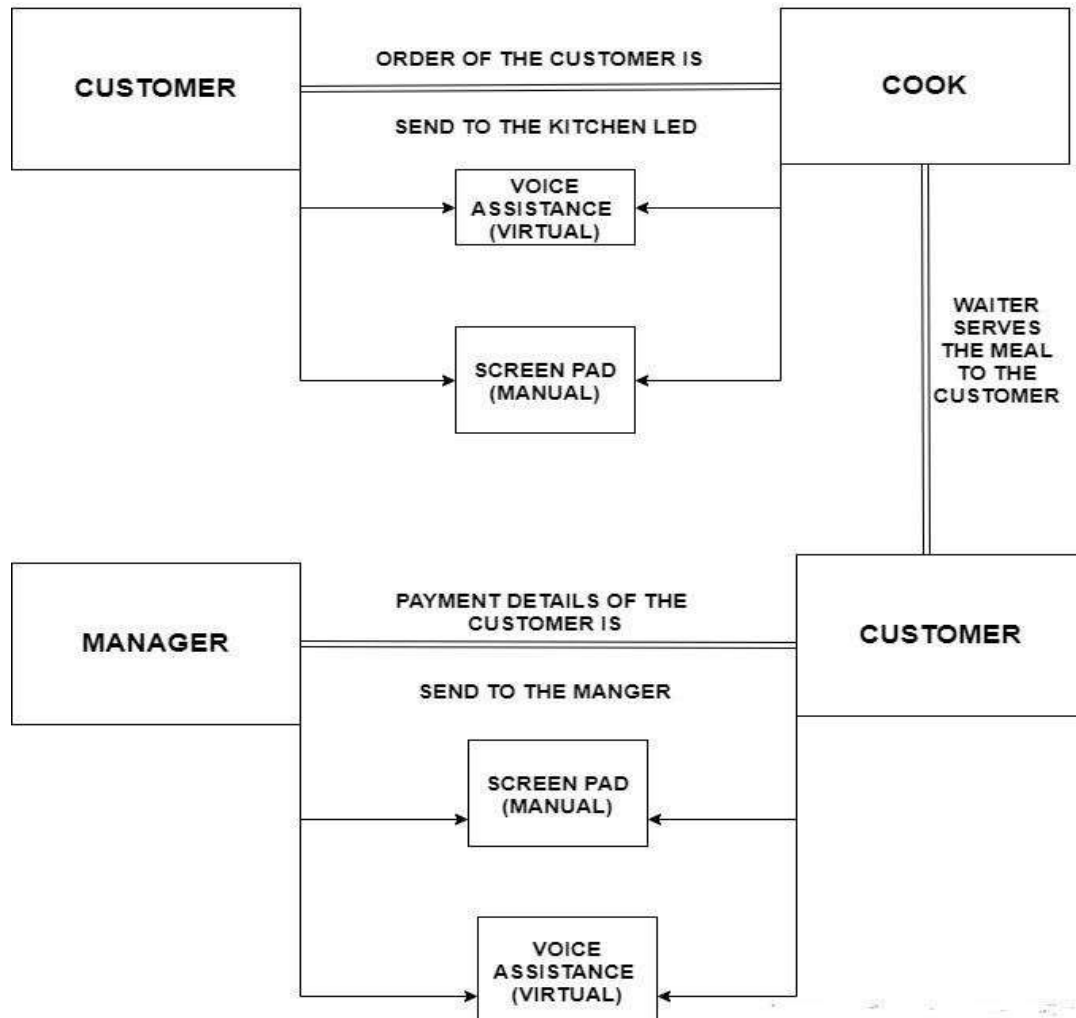
Database (Simulated) :

1. For simplicity, a **JSON** or **local Storage** system can be used to simulate a database to store orders, cart items, and customer preferences.

Security Considerations :

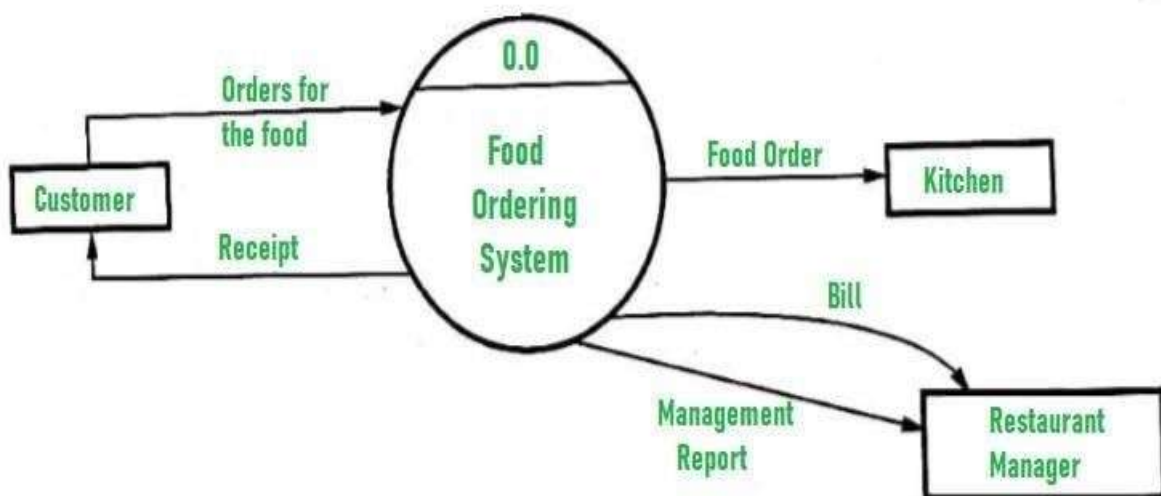
1. Although the system does not integrate a real payment gateway, proper security measures for handling sensitive data (e.g., passwords, payment details) should be considered in a real-world deployment.
2. Use secure protocols like HTTPS for data transmission.

System architecture



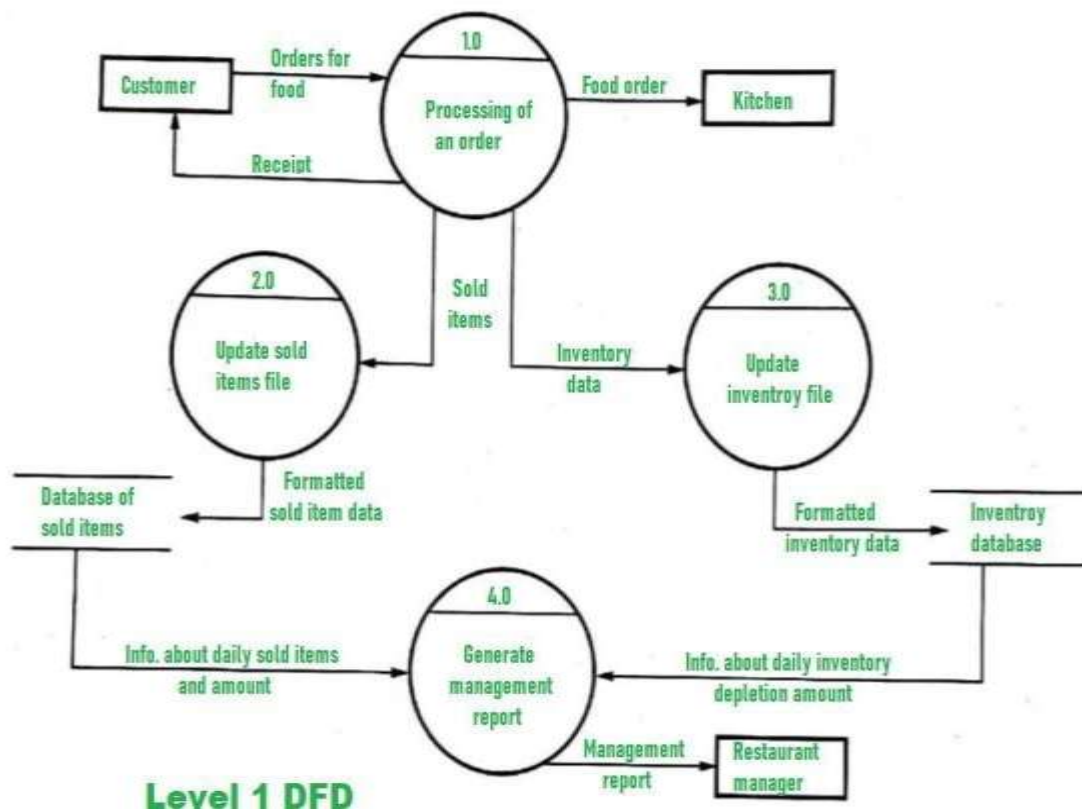
DFD

1. Level 0 DFD

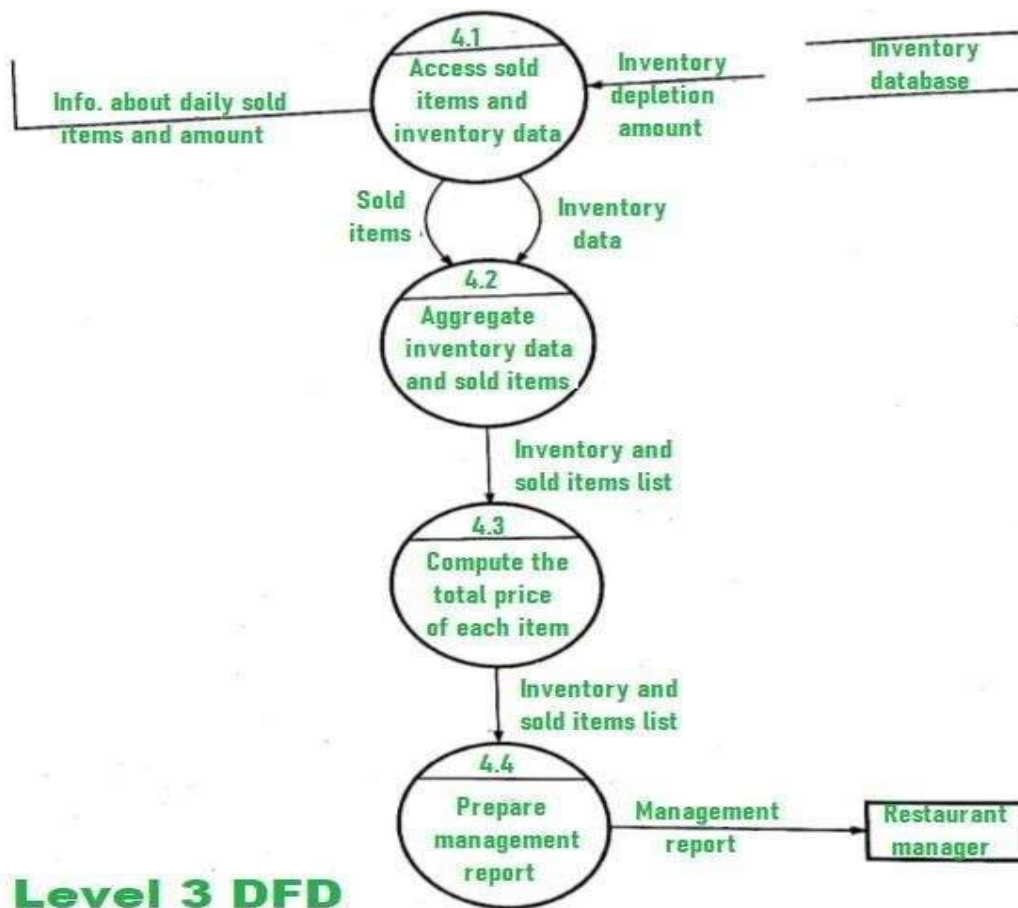


Level 0 DFD (Context Level)

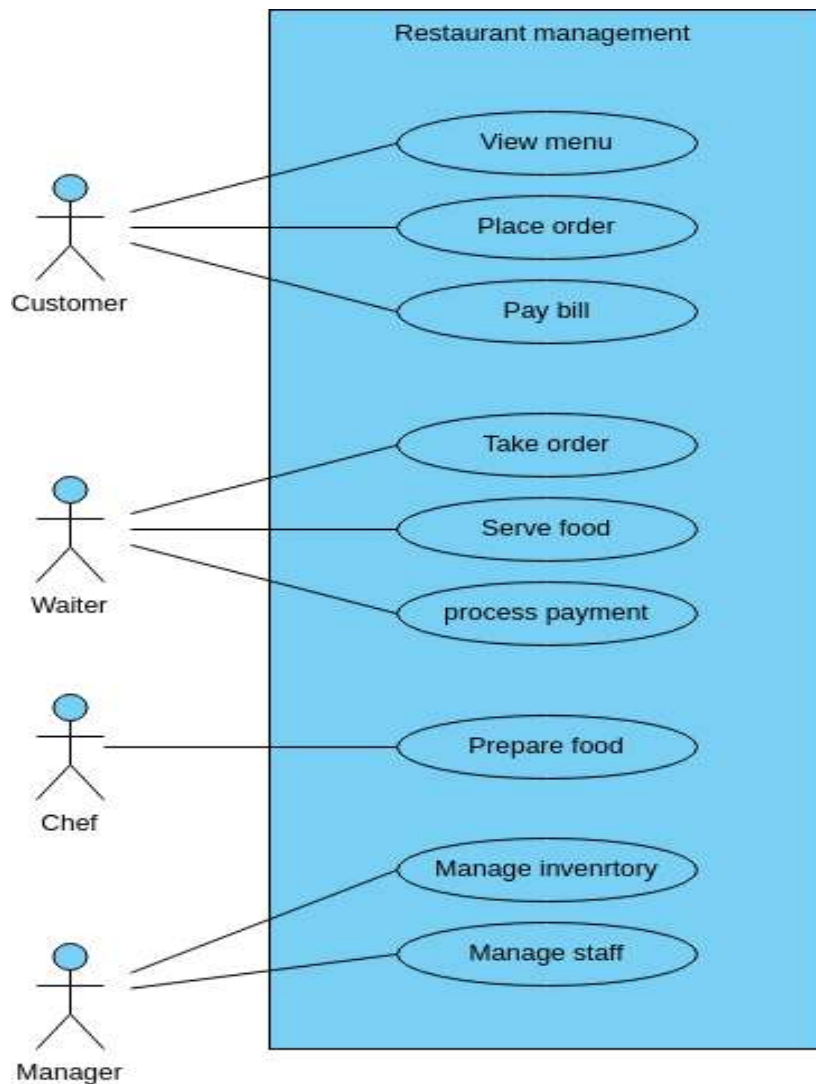
2. Level 1 DFD



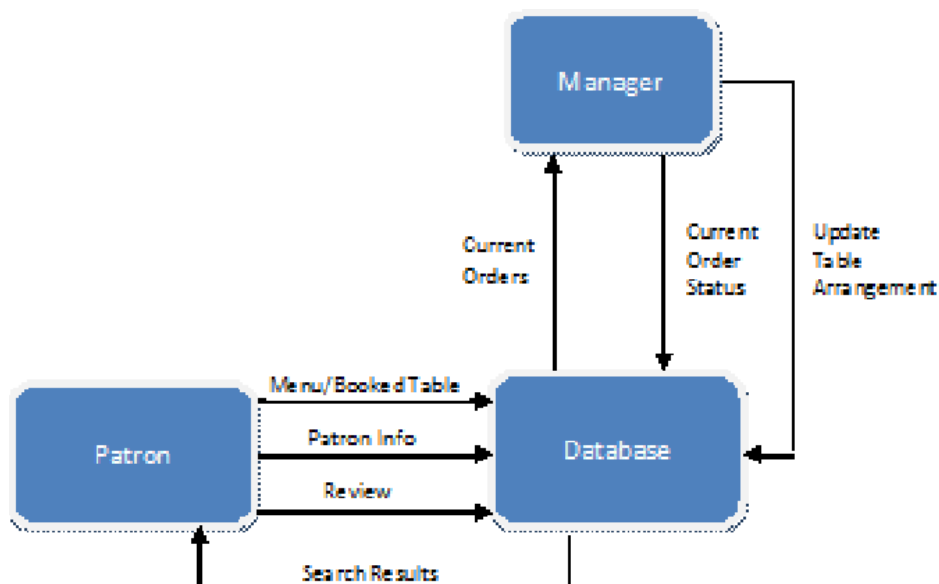
3. Level 2 DFD



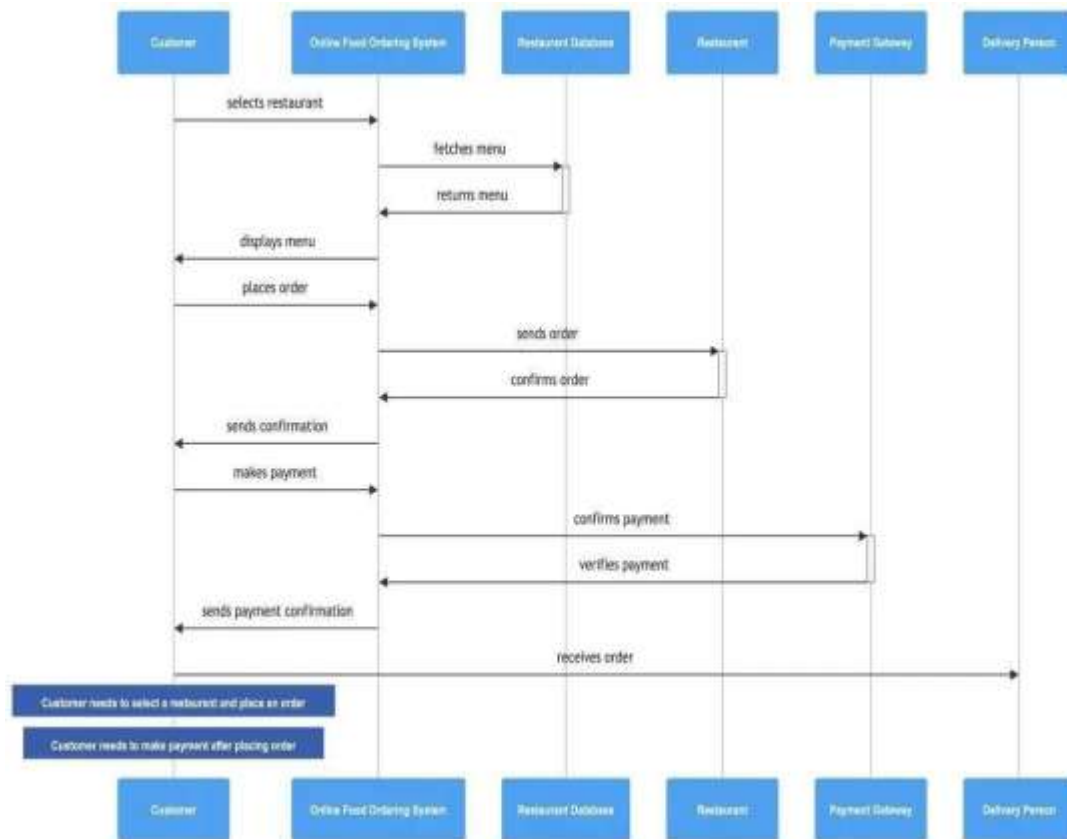
Use case diagram



Block diagram



Sequence diagram



Software requirement

Technology	Purpose
HTML	Structure and layout of web pages.
CSS	Styling and design of the web pages.
JavaScript	Dynamic content manipulation, event handling, and logic.
Git	Version control and collaboration.
VS Code/Sublime Text	Code editing and development.
Google Chrome/Firefox	Testing and debugging the application.

ACTION PLAN

Duration	Task/Action
15/07/24 - 31/07/24	Research - Search of Software and their features - Clearly define the goals and functionalities of the virtual mouse
01/08/24 - 15/08/24	Finalize the Project - Define the core features and scope - Set project goals and objectives - Create a clear project timeline
16/08/24 - 30/08/24	Requirement Analysis - Identify detailed functional requirements - Prioritize features and tasks

	- Define user interactions and flows
01/09/24 - 15/09/24	Planning - Create a detailed system architecture - Break down features into modules - Plan for the packages of project
16/09/24 - 15/10/24	System and Software Design - Design interface and user experience - Setup development environment for css,js,html. - Develop a simple program for tracks a order food
16/10/24 - 20/10/24	Building Project Units - Implement online order in tablet - Add basic needs of order. - Implement resource of insert table no. - implement online payment method.
21/10/24 - 31/10/24	Coding on Different Units - Code for activating required packages - Implement CSS,JS,HTML - Develop payment different method to payment. - Code for dining ordering system.
01/11/24 - 12/11/24	Testing of Particular Units - Test each unit for functionality - Debug and fix issues - Optimize performance for all movements - Conduct integration testing - Enhance the performance of software

IV. REFERENCES

"Android Programming"

- [1] Author: R. S. Salaria
- [2] Topic Covered: Android application development, user interface design, database management, and integrating features into Android apps.
- [3] Description: This book provides a detailed introduction to Android programming, covering fundamental concepts, coding practices, and essential app functionalities, which is perfect for building a hotel dining app. The book is practical and designed for beginners and intermediate learners.
- [4] Link: <https://www.mypustak.com/>

"Mobile App Development with React Native"

- [5] Author: Ravindra Babu
- [6] Topic Covered: React Native for building cross-platform mobile apps, UI design, mobile app performance, and integration of APIs.
- [7] Description: This book focuses on using React Native, a popular framework for building apps for both Android and iOS with a single codebase. It will be useful if you're looking for cross-platform development for your dining app.
- [8] Link: <https://www.freelancer.es/>

"Mobile App Security Testing"

[9] Author: S. K. Soni

[10] Topic Covered: Mobile app security, data encryption, secure user authentication, and secure communication.

[11] Description: Security is a critical aspect of any application, particularly when dealing with user data. This book covers how to secure mobile apps, including Android/iOS apps, through encryption, authentication.

[12] Link: <https://www.ibm.com/>

Biography

[13] Mobile app development (Android/iOS)

[14] UI/UX design and prototyping

[15] Database management and backend development

[16] Payment gateway integration

[17] Problem-solving and debugging

[18] Project documentation and reporting

[19] Presentation and communication skills